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LiDAR–RADAR Sensor Fusion for 3D Obstacle Detection and Multi-Object Tracking (Phase 2)

1. Introduction and Project Overview

Environmental perception is an essential component of autonomous driving, allowing vehicles to detect surrounding objects, estimate their positions, and understand their movements.

In Phase 1, a LiDAR-based obstacle detection system was developed using the nuScenes Mini dataset. The algorithm applied RANSAC for ground removal, DBSCAN for point-cloud clustering, and 3D bounding boxes to represent potential obstacles. Although this approach successfully detected geometric structures, it was limited to individual LiDAR scans and did not estimate object motion or maintain identities between consecutive frames.

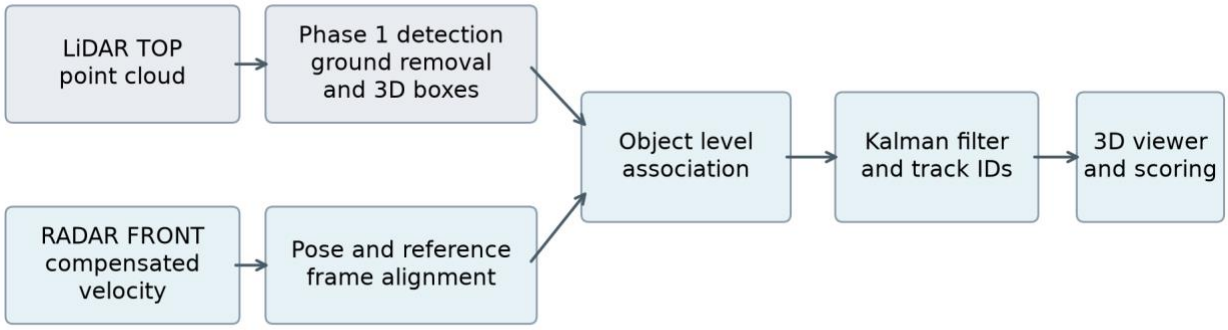
Phase 2 extends the previous system by introducing RADAR sensor fusion and multi-object tracking. LiDAR provides spatial information about potential obstacles, while RADAR contributes velocity measurements. These measurements are combined using a Kalman filter to estimate object positions and velocities over time.

The main objectives of Phase 2 are to:

1) Integrate LiDAR and RADAR measurements using a common coordinate system.
2) Associate RADAR returns with LiDAR object detections.
3) Implement Kalman filtering and persistent object tracking.
4) Evaluate the contribution of RADAR by comparing LiDAR-only and LiDAR–RADAR tracking.
5) Analyze detection limitations and identify improvements for Phase 3.

The project continues using the nuScenes Mini dataset described in Phase 1, consisting of 404 annotated samples across 10 driving scenes. The implementation uses the LIDAR_TOP and RADAR_FRONT sensors.

- **Phase 2 system architecture**



2. Methodology

2.1 LiDAR and RADAR Processing

The LiDAR detection pipeline developed in Phase 1 remains the foundation of the system. LiDAR point clouds are transformed into the vehicle coordinate system, followed by RANSAC ground removal and DBSCAN clustering.

For Phase 2, the clustering parameters were adjusted to a neighborhood radius of 0.40 m and a minimum of eight samples. Geometric filters were also retained and extended to reduce unwanted detections, including unsuitable bounding-box dimensions, near-vehicle artifacts, and certain elevated horizontal fragments. A ground-model check was introduced to reject implausible road planes.

The RADAR_FRONT measurements are processed separately. RADAR positions and compensated velocity vectors are transformed into the LiDAR reference coordinate system using the sensor calibration and ego-vehicle poses.

Because the two sensors acquire data at slightly different timestamps, their time difference is calculated. The system accounts for differences in sensor and vehicle poses but does not explicitly compensate for the motion of individual targets between acquisition times.

- **LiDAR–RADAR Sensor Association**

After generating LiDAR detections, RADAR measurements are associated with nearby objects using their spatial positions.

Each LiDAR bounding box is expanded by a margin of 1.0 m in the horizontal plane. RADAR returns falling inside an expanded box are considered for association. If a RADAR return corresponds to multiple boxes, it is assigned to the closest eligible detection center.

For detections containing multiple associated RADAR returns, the compensated velocity vectors are averaged to obtain a representative velocity measurement.

This process allows LiDAR to provide object geometry and position while RADAR supplies additional motion information. Objects without associated RADAR measurements are still processed using LiDAR alone.

2.3 Kalman Filter and Multi-Object Tracking

A Kalman filter is implemented to estimate the horizontal position and velocity of detected objects across consecutive scans.

The state vector is defined as:

$$\mathbf{x} = \left[x,\ y,\ v_x,\ v_y\right]^T$$

where **(x)** and **(y)** represent the object's position, while **(v_x)** and **(v_y)** represent its velocity components.

The filter operates in two stages. During prediction, the next position is estimated using a constant-velocity motion model. During correction, LiDAR measurements update the estimated position, while associated RADAR measurements update the estimated velocity.

A multi-object tracker assigns an identification number to each detected object. Detections are associated with existing tracks using their predicted positions, with a maximum matching distance of 3 m.

When an object cannot be detected in a frame, its position can temporarily be predicted using the Kalman filter. Tracks are removed after exceeding two consecutive missed updates.

This enables the system to maintain object identities and estimate motion across a driving sequence rather than treating every scan independently.

3. Results and Performance Evaluation

The final system was evaluated using all 404 annotated samples from the nuScenes Mini dataset.

Two evaluations were performed: one measuring geometric detection performance against ground-truth annotations, and another comparing the tracking performance of LiDAR-only and LiDAR–RADAR configurations.

3.1 Object Detection Performance

The detection algorithm was evaluated against annotated vehicles and pedestrians within a 25m range.

A simplified bird's-eye-view Intersection over Union (IoU) threshold of 0.1 was used to associate predicted boxes with ground-truth objects.

Evaluation metric	Result
True positives	1,812
False positives	5,751
False negatives	4,222
Precision	24.0%
Recall	30.0%
F1-score	26.7%
Mean matched BEV IoU	0.378

The results demonstrate that the algorithm can detect and geometrically represent a portion of the surrounding objects. However, the relatively low precision and recall indicate a considerable number of incorrect detections and missed objects.

These limitations are mainly associated with sparse LiDAR measurements, clustering errors, and the difficulty of separating nearby objects using geometric methods alone.

The evaluation is a custom class-agnostic diagnostic rather than the official nuScenes detection benchmark.

3.2 LiDAR-Only VS LiDAR–RADAR Tracking

To evaluate the contribution of RADAR, two tracking configurations were compared using the same recorded detection inputs.

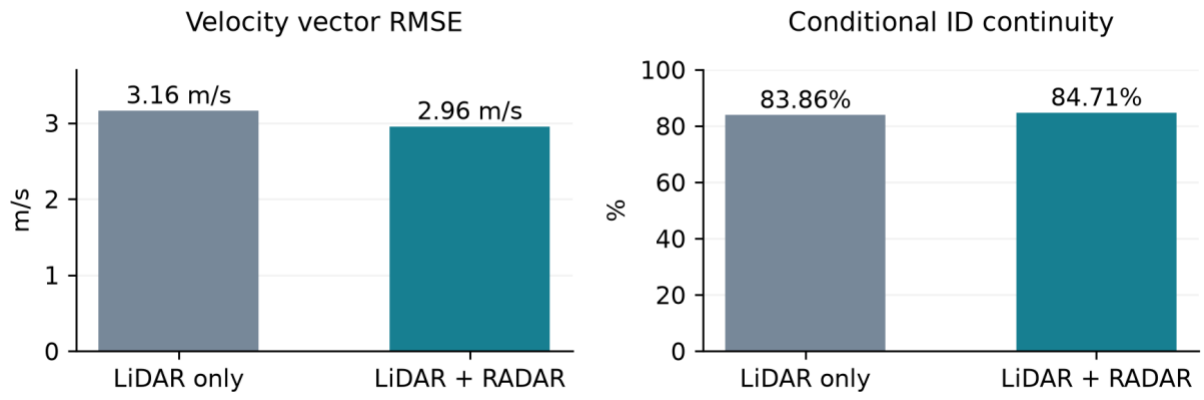
The first configuration used LiDAR position measurements and Kalman motion prediction without RADAR velocity updates. The second incorporated associated RADAR velocities for initialization and correction.

Both configurations used the same detection positions, tracking parameters, and evaluation procedure.

Table 2: LiDAR-only and LiDAR–RADAR tracking comparison

Tracking metric	LiDAR only	LiDAR + RADAR
Matched target observations	2,633	2,638
False track observations	9,265	9,214
Tracking precision	22.13%	22.26%
Tracking recall	43.64%	43.72%
Tracking F1-score	29.37%	29.50%
Conditional ID continuity	83.86%	84.71%
Consecutive ID changes	354	337
Mean matched position error	0.771 m	0.777 m
Matched velocity RMSE	3.163 m/s	2.956 m/s

Figure 2: Comparison of velocity RMSE and conditional ID continuity



The results show that introducing RADAR velocity information reduced the matched velocity root mean square error (RMSE) from 3.163 m/s to 2.956 m/s, representing an improvement of approximately **6.5%**.

Conditional track identity continuity also increased from 83.86% to 84.71%, while consecutive identity changes decreased from 354 to 337.

However, the improvements in tracking precision and recall were relatively small. Mean matched position error also increased slightly from 0.771 m to 0.777 m.

These findings indicate that RADAR provides a measurable but limited benefit to velocity estimation and identity continuity. The velocity and position errors are calculated only for matched tracks, and the two configurations do not necessarily match exactly the same objects.

3.3 RADAR Association Coverage

An additional analysis examined how frequently RADAR measurements were successfully associated with LiDAR detections.

Within the evaluated 25 m range, 718 out of 7,563 detection observations received associated RADAR returns, corresponding to approximately **9.5%**.

This limited association rate helps explain why the overall tracking improvement was modest.

Since the system uses only the front RADAR channel, many objects do not have usable associated RADAR measurements. Additionally, the simple spatial association method may struggle when objects are close together or when RADAR returns fall outside the expanded LiDAR bounding boxes.

Therefore, improving sensor coverage and measurement association represents an important direction for future development.

4. Visualization and Limitations

4.1 Interactive 3D Visualization

An interactive 3D viewer was developed to visualize the final perception and tracking results.

Unlike the static visualizations used in Phase 1, the new interface allows consecutive scans to be played back, displaying changes in detected objects and their estimated motion.

The viewer displays LiDAR point clouds, RADAR returns, object bounding boxes, track identification numbers, estimated velocities, and tracking trajectories.

Detections with associated RADAR information are visually distinguished from LiDAR-only detections. The interface also allows ground-truth bounding boxes to be displayed for comparison.

Users can rotate and zoom the scene, switch between top-down and 3D views, navigate between scans, select tracked objects, and export screenshots.

Figure 3: Interactive 3D visualization of LiDAR–RADAR fusion and object tracking



This visualization provides an effective method for examining the algorithm's behavior and identifying detection and tracking errors.

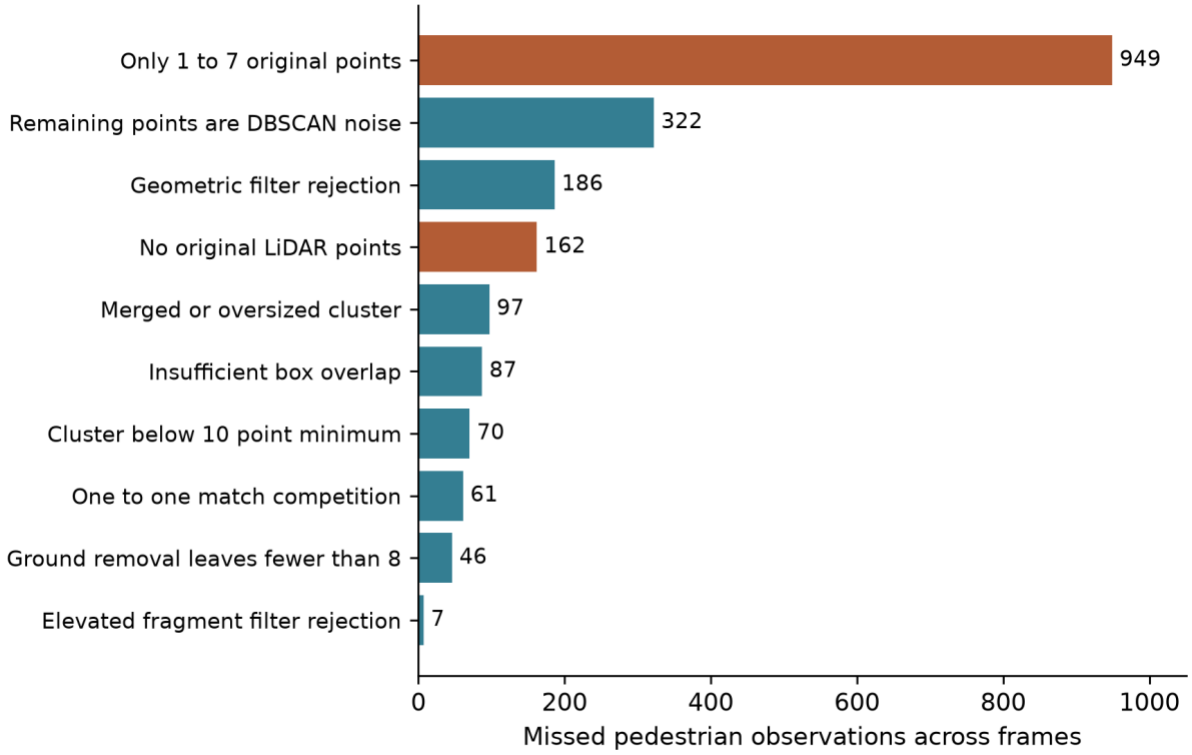
Visual element	Meaning
Green to blue point colors	Distance-based LiDAR coloring, not predicted classes
Yellow and green boxes	Detections without and with associated RADAR
Pink points	RADAR returns
Track labels and arrows	IDs, Kalman speed estimates and one-second motion arrows
Optional gray crosses	Active predictions without a current detection
Optional dashed white boxes	Ground-truth reference annotations

4.2 Detection Limitations and Pedestrian Analysis

Despite implementing sensor fusion and tracking, the system remains limited by the accuracy of the original geometric detection process.The full-dataset evaluation included 2,530 pedestrian observations, of which 543 were successfully matched and 1,987 were missed.

A diagnostic analysis was performed to identify the main stages contributing to missed pedestrian detections.

Figure 4: Main pedestrian detection failure categories



The largest limitation was insufficient LiDAR point support. A total of 1,111 missed pedestrian observations contained fewer than eight original LiDAR points inside their annotated 3D bounding boxes.

This demonstrates that smaller or distant objects can be difficult to detect reliably using a single LiDAR scan.

Another core limitation is object merging. When pedestrians stand close to vehicles or surrounding structures, DBSCAN may combine their points into a single large cluster. This can prevent the algorithm from generating separate bounding boxes for each object.

Furthermore, geometric filtering may reject some valid objects when only part of their surfaces is visible.

These results show that parameter tuning alone is insufficient to solve every detection problem. Improving detection reliability requires better handling of sparse observations, object separation, and semantic information.

The diagnostic categories identify the first observed failure stage rather than proving a unique physical cause in every case.

5. Future Work (Phase 3)

Phase 2 successfully introduced RADAR-assisted tracking, but the experimental results identified several limitations that will guide the next stage.

The main objective of Phase 3 will be to improve object detection and tracking reliability, particularly for pedestrians and objects that produce sparse or ambiguous sensor measurements.

The proposed improvements include:

1. Improved object detection and classification

A pretrained 3D object detection model could be introduced to distinguish vehicles, pedestrians, and other road users. This would extend the current geometry-based detector, which produces candidate objects without semantic class predictions.

2. Enhanced LiDAR–RADAR fusion

The remaining RADAR sensors could be incorporated to increase spatial coverage and improve the availability of velocity measurements. More advanced association techniques could also reduce incorrect sensor matches.

3. Improved clustering and object separation

Range-dependent clustering or more advanced segmentation methods could help separate nearby objects while preserving sparse detections.

4. More reliable tracking and evaluation

Future development could investigate stronger data association methods and motion models to reduce identity changes and false tracks. Evaluation should also include independent validation data and established object detection and tracking benchmark protocols.

These improvements would be evaluated individually to determine which changes provide measurable benefits.

6. Conclusion

Phase 2 extends the original LiDAR-based 3D obstacle detection system into a multi-sensor perception and tracking prototype.

The developed system combines geometric LiDAR detections with RADAR velocity measurements and uses Kalman filtering to estimate object motion across consecutive frames. An interactive 3D viewer was also implemented to visualize detected objects, sensor measurements, and tracking results.

The experimental evaluation demonstrated that RADAR-assisted tracking reduced matched velocity RMSE by approximately **6.5%** and improved conditional identity continuity by 0.85 percentage points compared with LiDAR-only tracking.

However, the system continues to experience false detections, missed pedestrians, merged object clusters, and limited RADAR association coverage.

Overall, Phase 2 demonstrates the feasibility of classical LiDAR–RADAR sensor fusion for object tracking while identifying the main challenges that must be addressed in Phase 3 to achieve more reliable autonomous vehicle perception.

References

[1] Caesar, H., et al. (2020). *nuScenes: A Multimodal Dataset for Autonomous Driving*. Proceedings of CVPR. https://arxiv.org/abs/1903.11027

[2] nuScenes development kit. Point-cloud data classes and sensor conventions. Used package version 1.2.0. https://github.com/nutonomy/nuscenes-devkit